

KHAMAR-DABAN, Russian FederationWMO: **308150**Lat: **51.5297N**Lon: **103.5890E**Elev: **4731**StdP: **12.35**Time Zone: **8.00 (E08)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-17.0	-23.2	1.8	-20.9	-19.6	2.2	-16.4	15.1	10.0	11.3	9.7	2.0	30	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	73.7	57.9	70.5	57.0	67.5	55.6	60.9	68.2	59.1	66.8	57.6	64.4	5.5	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.1	86.1	63.9	56.5	81.1	61.7	54.9	76.5	60.1	29.4	68.4	28.0	67.1	26.9	64.5	66.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	9.4	7.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	-25.8	64.2	6.1	2.0	-30.2	65.6	-33.8	66.8	-37.2	68.0	-41.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.3	3.7	5.3	17.2	28.8	39.7	52.1	56.2	53.8	42.7	28.8	13.7	8.0
	DBStd	21.05	10.35	10.96	11.64	9.61	9.24	7.66	5.26	7.03	9.57	9.99	12.34	10.11
	HDD50	8104	1435	1251	1017	639	340	66	8	38	260	658	1090	1302
	HDD65	13042	1900	1671	1482	1086	784	391	276	353	670	1122	1540	1767
	CDD50	550	0	0	0	3	21	128	201	156	40	1	0	0
	CDD65	16	0	0	0	0	1	4	4	6	1	0	0	0
	CDH74	75	0	0	0	0	6	20	15	27	7	0	0	0
	CDH80	3	0	0	0	0	0	1	0	2	0	0	0	0
	WSAvg	3.2	2.7	2.6	3.2	3.5	3.4	3.5	3.3	3.4	3.6	3.2	2.7	2.6
	PrecAvg	55.2	1.2	1.1	2.2	3.8	4.4	7.3	12.0	9.1	6.3	3.3	2.4	1.7
	PrecMax	78.2	2.1	2.2	3.7	7.2	8.7	19.7	28.6	20.5	16.1	7.1	4.4	3.9
(15) Precipitation	PrecMin	43.4	0.4	0.2	1.0	1.8	1.2	1.8	3.8	3.6	2.6	1.3	0.7	0.0
	PrecStd	8.3	0.5	0.5	0.7	1.1	1.5	3.6	5.7	4.3	2.5	1.3	0.9	0.7
	0.4%	DB	30.9	34.4	49.6	61.2	72.4	76.8	75.9	77.3	73.8	59.0	46.7	35.5
	MCWB	23.9	26.3	36.6	42.4	51.1	59.0	62.1	59.7	53.6	42.1	34.7	28.0	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	26.1	29.9	44.4	53.9	65.0	72.4	72.5	72.6	66.3	51.9	40.5	29.7
	MCWB	20.4	23.3	33.2	39.0	47.9	56.2	59.5	58.2	50.3	38.8	31.8	23.6	
	5%	DB	21.8	25.6	39.7	48.5	60.1	68.5	69.8	68.7	61.3	47.6	35.1	24.9
	MCWB	17.7	20.2	30.4	36.1	45.4	54.5	58.9	57.1	49.9	36.3	28.2	20.8	
	10%	DB	17.6	21.1	35.0	44.4	56.0	64.9	66.7	65.5	57.7	43.7	30.2	21.3
	MCWB	14.3	16.9	27.4	34.3	43.3	53.1	57.8	55.7	47.8	34.1	25.0	18.2	
	0.4%	WB	23.9	26.7	37.1	43.9	53.1	61.1	64.7	62.4	56.4	42.7	35.3	28.4
	MCDB	30.2	33.3	48.3	58.2	70.0	70.7	74.0	73.2	64.6	56.9	45.0	34.9	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	20.6	23.3	33.4	40.0	49.1	58.3	61.8	59.9	53.3	39.7	32.0	24.1
	MCDB	25.5	29.3	43.9	52.1	63.0	67.6	68.2	68.3	62.3	50.8	39.8	28.8	
	5%	WB	17.7	20.4	30.5	37.0	46.5	56.5	60.0	58.3	50.9	36.8	28.7	21.1
	MCDB	21.4	25.3	39.2	47.2	58.3	64.5	66.6	66.2	60.1	46.1	34.7	24.8	
	10%	WB	14.5	17.0	27.6	34.6	43.7	54.8	58.3	56.7	48.5	34.6	25.4	18.1
	MCDB	17.2	21.3	34.6	43.8	54.8	62.1	64.7	63.7	57.0	42.5	30.1	21.2	
	5% DB	MDBR	15.9	16.6	17.9	17.1	17.7	15.4	14.2	13.5	15.3	15.3	15.9	14.8
	MCDBR	19.5	19.1	22.5	23.5	24.5	19.6	18.9	18.5	19.4	20.8	19.8	18.1	
(36) Mean Daily Temperature Range	MCWBR	14.6	14.0	13.6	12.6	12.3	9.9	10.4	10.2	10.2	11.6	12.7	13.3	
	5% WB	MCDDBR	20.0	19.3	21.9	21.5	22.6	16.1	15.9	16.0	17.8	19.5	19.1	17.2
	MCWBR	15.2	14.1	13.5	11.9	12.1	9.3	9.9	9.6	10.0	11.7	12.3	12.6	
	taub	0.233	0.257	0.295	0.337	0.381	0.373	0.369	0.342	0.298	0.280	0.254	0.235	
(40) Clear-Sky Solar Irradiance	taud	2.232	2.213	2.205	2.299	2.218	2.355	2.401	2.461	2.546	2.498	2.310	2.190	
	Ebn at Noon	254	278	286	286	276	279	279	282	286	270	243	229	
	Edh at Noon	21	29	36	37	42	37	35	31	25	22	19	18	
	RadAvg	440	783	1231	1566	1720	1734	1616	1408	1131	742	446	317	
(44) All-Sky Solar Radiation	RadStd	17	30	57	80	82	111	120	94	97	43	25	16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A